WELLSLING · PROJECT PLAN

WellSling project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Concept choice (D15, option A) and D16 (a) decided by Amish on 2026-10-25
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade)

WellSling project plan

WellSling is at TRL 3 and stage P3 (design outputs): model, drawings, priced BOM, build plan, decisions register, design FMEA and traceability are in place for option A, and every concept option is modelled at detail level D3. The next gate, P4 (verification), is on hold under the portfolio cap. The decisions requested of Amish are the concept choice (WSL-DDR-003, option A recommended) and review of the five fixes made under his standing instruction (WSL-DEC-001 D10 to D14). Method: STANDARDS sections 19 and 23.

> **Safety:** P4 work starts with a proof-load test of the built kit at 1.5 times the hook load by a competent person. Nothing is lifted over a well, and no animal is lifted, before that.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and users	Amish	WSL-PRB-001 with users, needs and stakeholders	Done (v0.3)
STEM check	Design lead	WSL-STM-001, `stem_check.py`, independent review	Done; open items need TRL 4
Concept options	Design lead	WSL-OPT-001, WSL-DDR-003, options models in `cad/src/model.py`	Done; option A chosen by Amish, 2026-10-25
Requirements and traceability	Design lead	WSL-URS-001, WSL-PRS-001, WSL-RTM-001	Done
Calculations	Design lead	WSL-CAL-001 v0.2	Done
Detailed CAD	Design lead	`cad/src/model.py` at D3 for options A, B and C; `cad/fidelity.json`; STEP per part and per assembly	Done
Drawings, BOM and build plan	Design lead	WSL-DWG-001, 101 to 116; `bom/bom.csv`; WSL-BLD-001	Done for option A
Regulatory, feasibility, risk, market	Design lead	WSL-REG-001, WSL-FEA-001, WSL-RSK-001, WSL-MKT-001	Done (first pass)
Renders and storefront	Coordinator	Cycles renders on Amish's Mac: options picture, product renders, cards	Scenes exported; rendering pending
Co-design partner	Amish	Agreement with an animal rescue organisation or a dairy cooperative union, with a veterinarian	Not started
Verification (P4)	Partner and lifting inspector	Proof-load test, brake test, dummy trials	On hold (stage cap P3)

Resources needed

Table 2. Resources (P4 is planned work, on hold).

Resource	Deliverable	Effort	Estimated cost (USD)
District fabrication workshop	One kit built to WSL-BLD-001	About 10 working days	3,093 (BOM estimate)
Competent lifting inspector	Proof-load test at 1.5 times the hook load, brake test	1 day	To assess
Veterinarian or veterinary university	Review of bands, pressure, lift time and handling	2 days	To assess
Animal rescue organisation or cooperative	Test pit, dummy, crew for timed drills and first-operator trials	3 days	To assess
Galvanising plant	Hot-dip galvanising of the made steel parts	Batch	Included in the BOM estimate

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	Problem, STEM check, concept options, first REG, FEA, PLN	Met
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT, REG complete	Met
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM and feasibility closed	Met, except five STEM items that need TRL 4 evidence ; the concept choice was made on 2026-10-25
P3 Design outputs	Model at D3, STEP, drawings, priced BOM, build plan, DEC, design FMEA, traceability	Met (2026-10-25)
P4 Verification	Proof-load and brake tests, dummy trials, timed drill, user trial, RSK updated	On hold
P5 Validation	Field trial with a partner at real wells, veterinary validation	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. Amish: confirm or reverse the fixes D10 to D14, D17 and D18, and answer the scope question on animals over 600 kg. The concept (option A) and the graft (keep the chains) were decided on 2026-10-25.
2. Amish: decide whether to graft option B's best idea, independent chest and flank adjustment, onto option A (WSL-DEC-001, open decision).
3. Coordinator: render the options picture and the product renders from the exported scenes, then the cards.
4. Amish: approach the first co-design candidates (WSL-PRB-001) when he chooses to start TRL 4.

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